

SHAPEIC: DESIGN-SPACE EXPLORATION

Hierarchical LUT-based exploration from electrical models to layout-aware design

Daniel Arevalos

AEMY Group

Automatic Analog Sizing

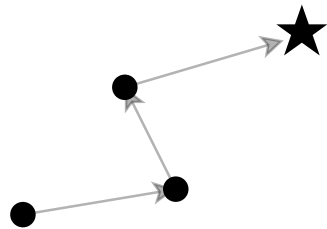


ShapelC: Design-Space Exploration – Hierarchical LUT-based exploration from electrical models to layout-aware design

Daniel Arevalos – AEMY Group

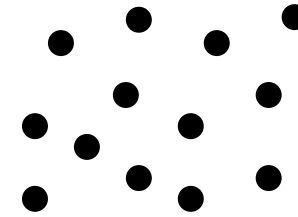
Automatic Analog Sizing

► Optimization-driven



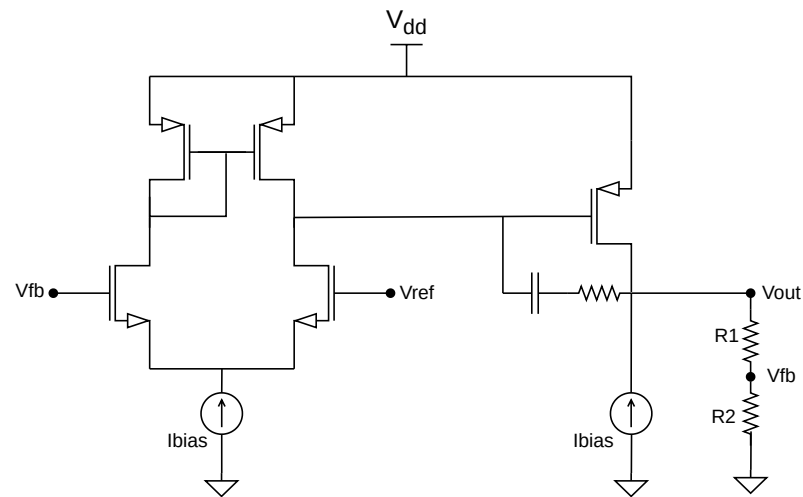
- ✓ Targeted evaluations
- ✓ Many algorithms available
- ✗ Partial View of the design space
- ✗ May converge to local optima

► Design-space exploration



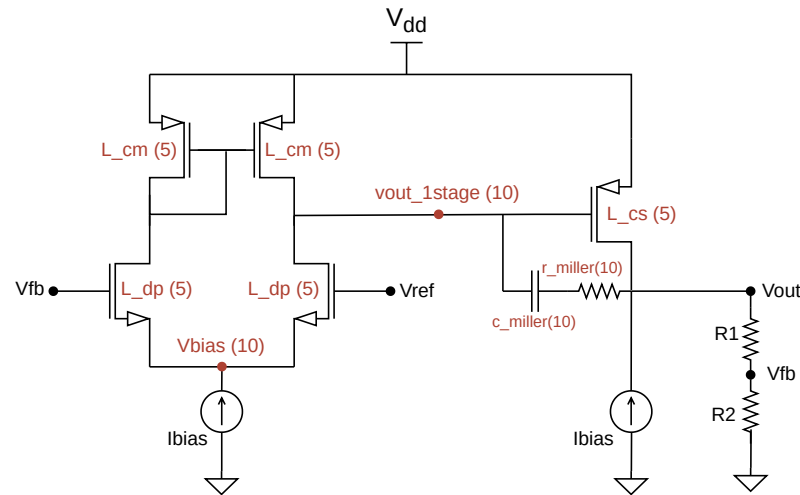
- ✓ Design insight
- ✓ Alternative solutions
- ✗ Expensive to build
- ✗ Scales poorly without abstraction

Let's explore a 2 Stage Amplifier



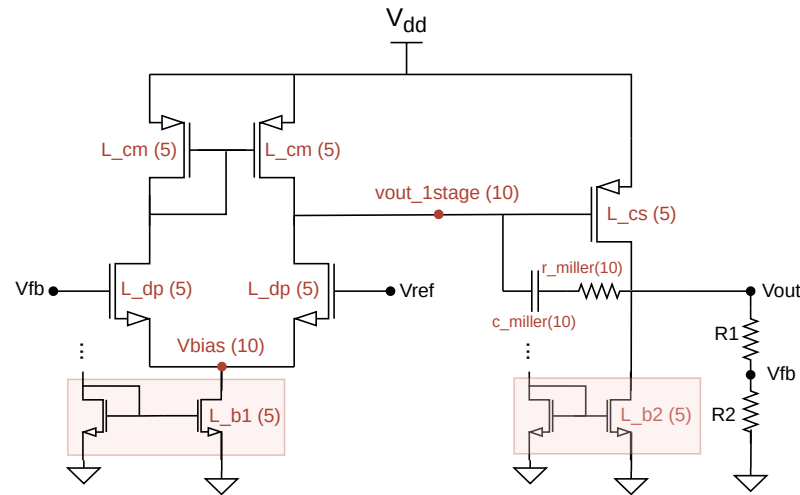
► Simple 5 transistor topology

Let's explore a 2 Stage Amplifier



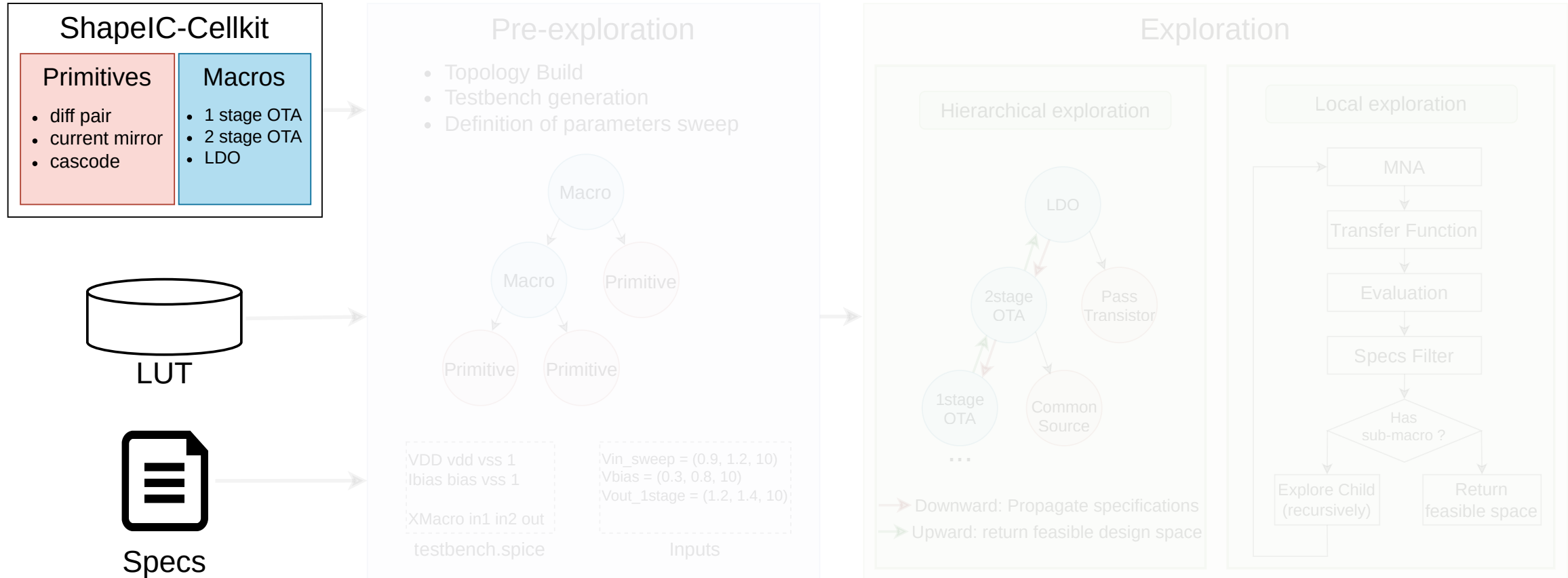
- ▶ Simple 5 transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: $\approx 2h$ (with 8 cores)
 - ▶ Equation evaluations: $\approx 2s$ (but you need to derive them)

Let's explore a 2 Stage Amplifier

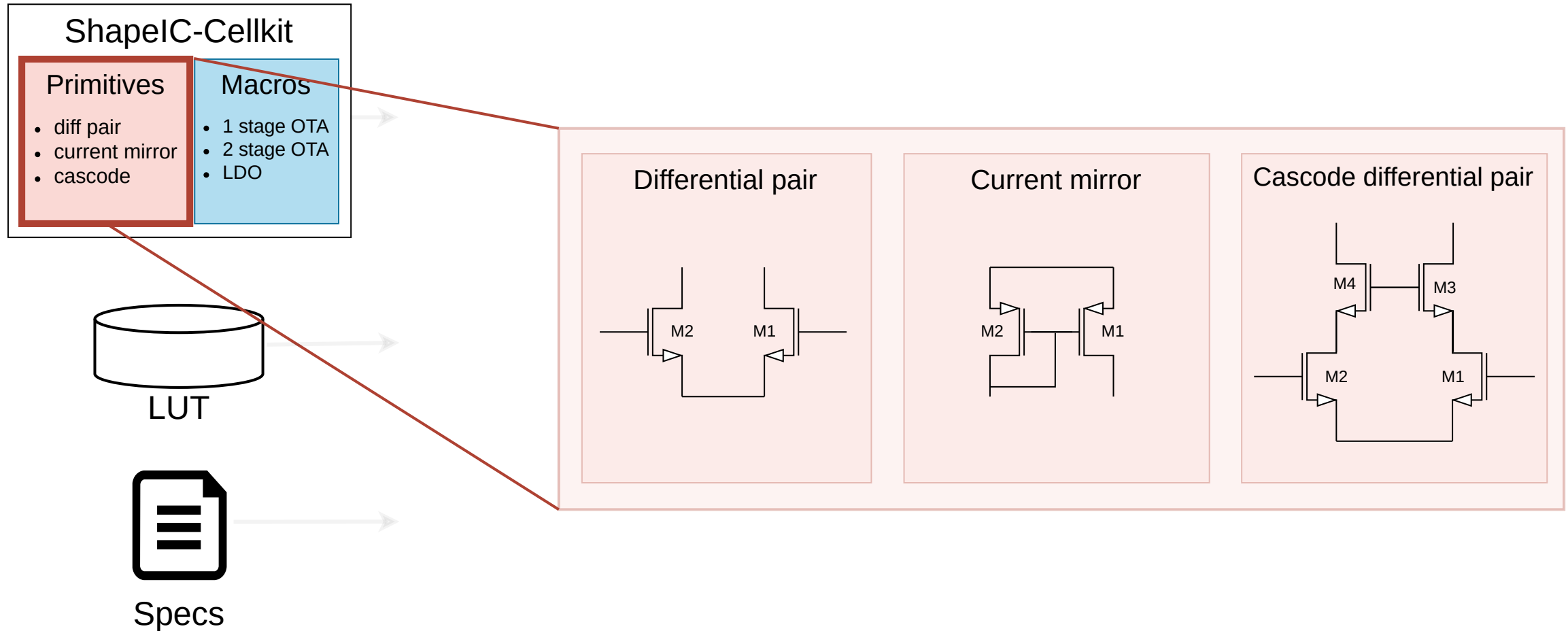


- ▶ Simple 5 transistor topology
- ▶ 1250000 design candidates
 - ▶ Simulation: $\approx 2h$ (with 8 cores)
 - ▶ Equation evaluations: $\approx 2s$ (but you need to derive them)
- ▶ 31250000 design candidates
 - ▶ Simulation: $\approx 1month$
 - ▶ Equation evaluations: $\approx 1min$ (but you need to recalculate equations)

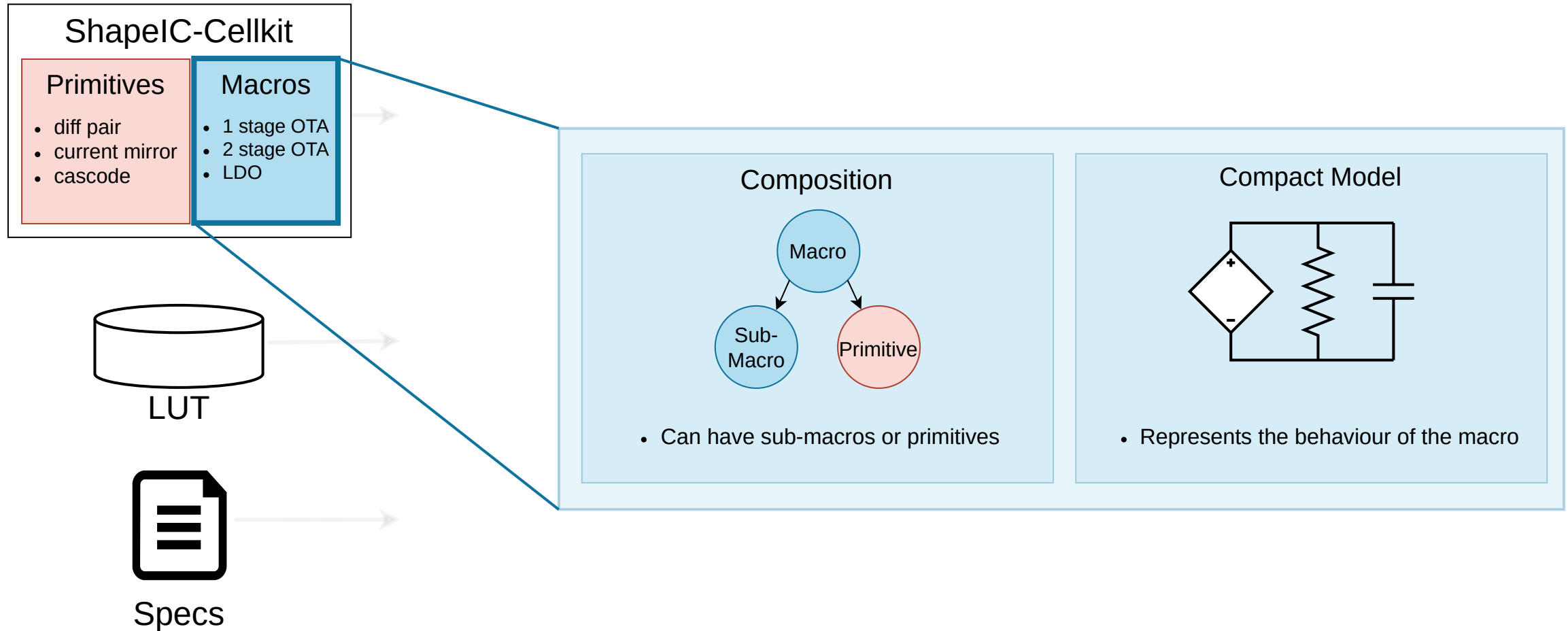
ShapelC: Building the Design Space



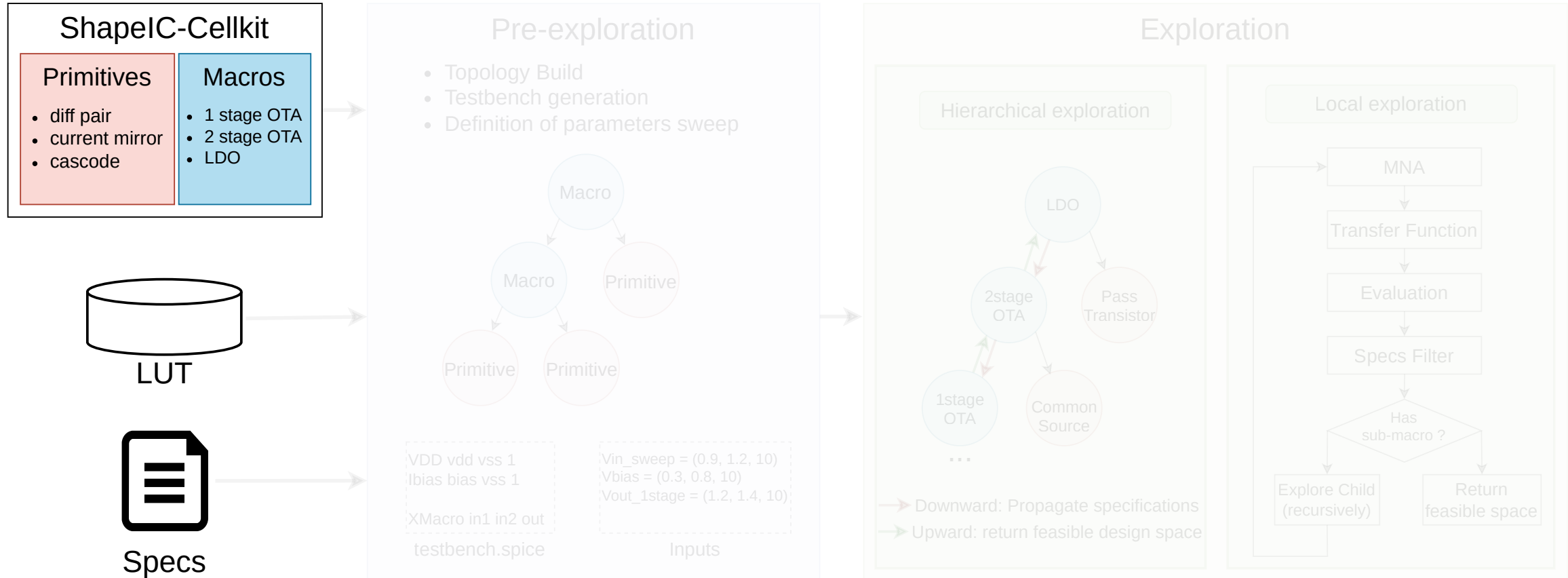
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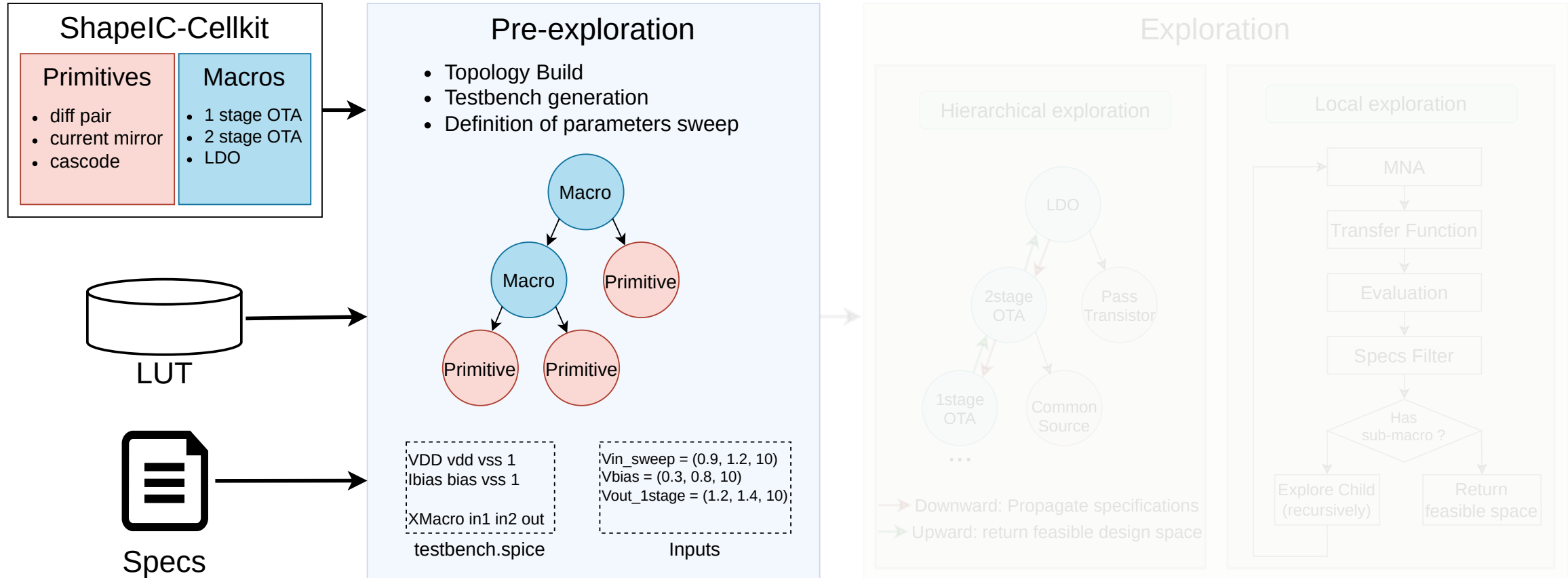
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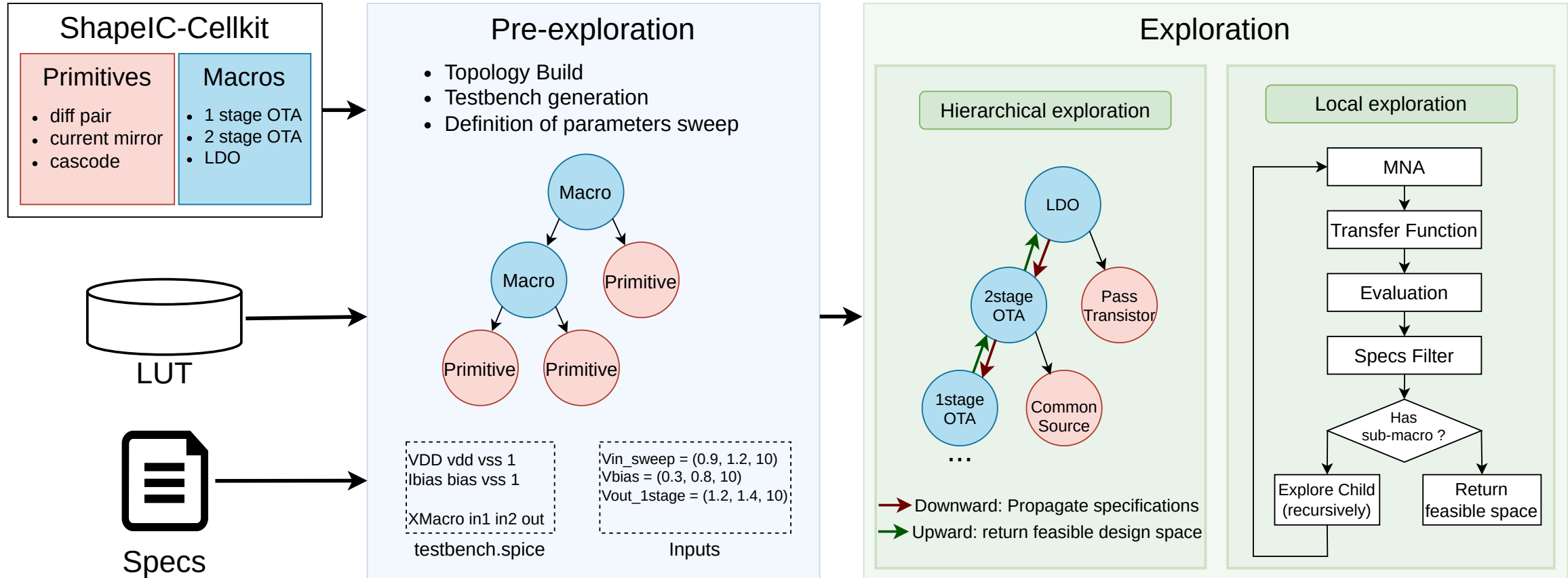
ShapelC: Building the Design Space



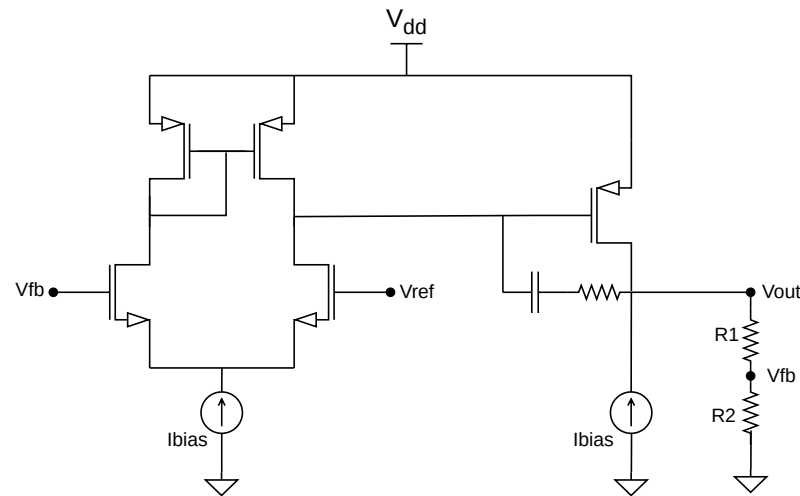
ShapelC: Building the Design Space



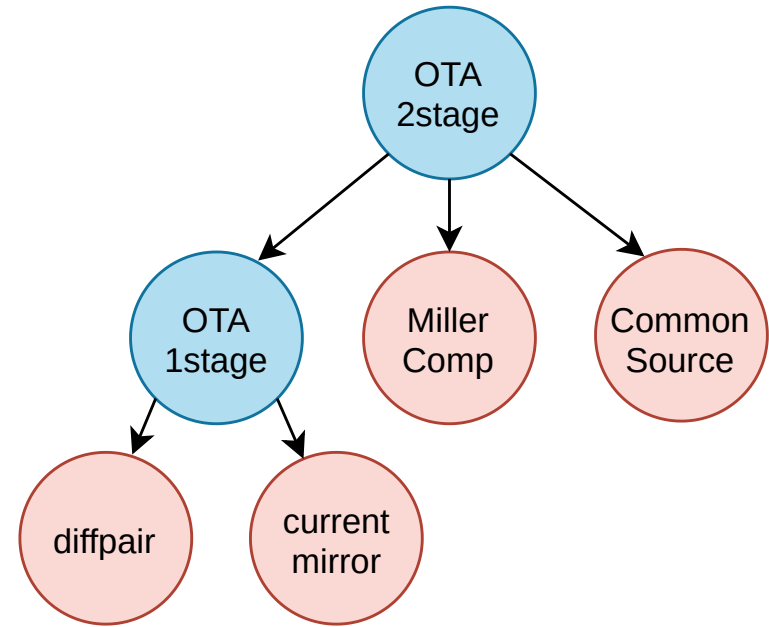
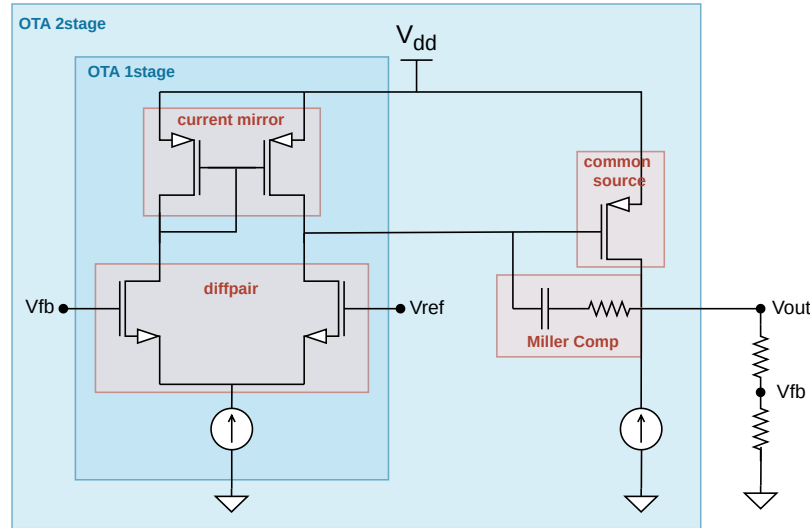
ShapelC: Building the Design Space



ShapelC in Action: 2 Stage Amplifier (OTA)



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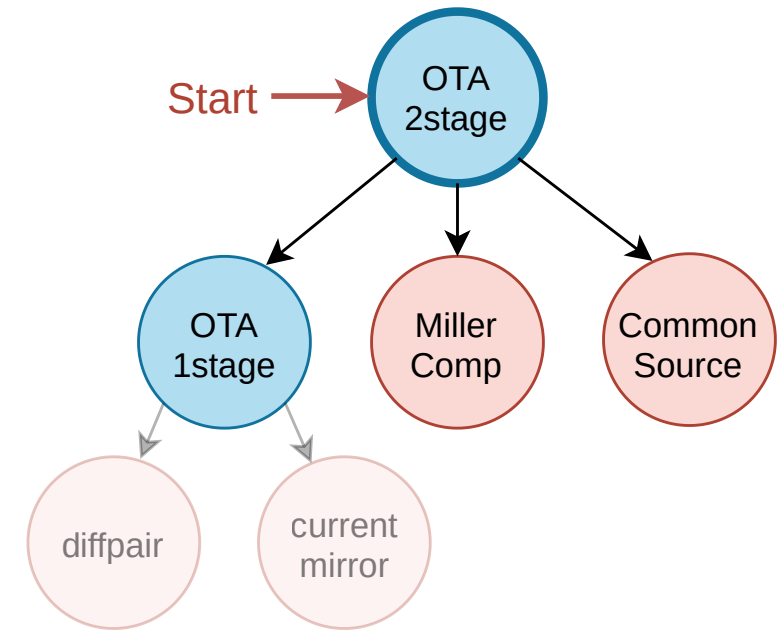


- ▶ Testbench: AC analysis
- ▶ Specs:
 $Gain \geq 70, BW \geq 10kHz, PM \geq 60$

ShapelC in Action: 2 Stage Amplifier (OTA)

1. Hierarchical exploration

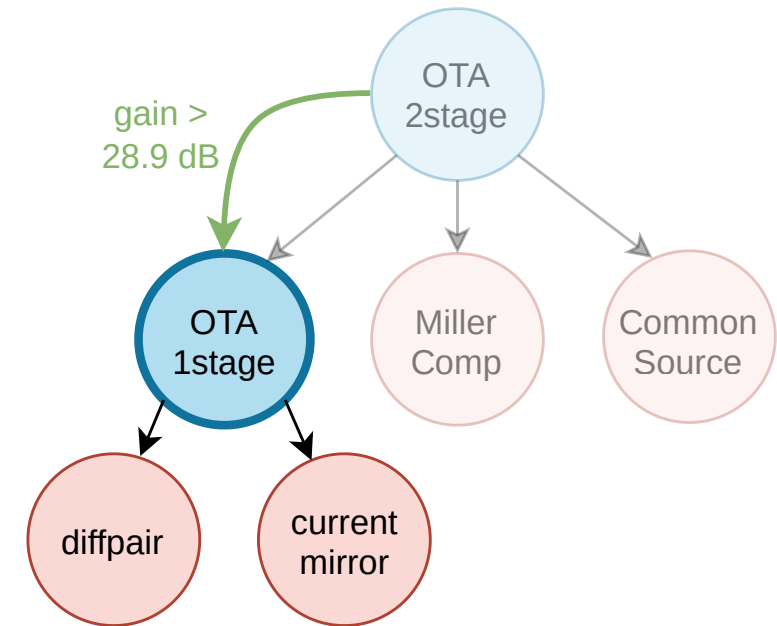
- ▶ TOP exploration: 500k → 2021
- ▶ Derived constraint: $\text{Gain} \geq 28.89 \text{ [dB]}$



ShapelC in Action: 2 Stage Amplifier (OTA)

1. Hierarchical exploration

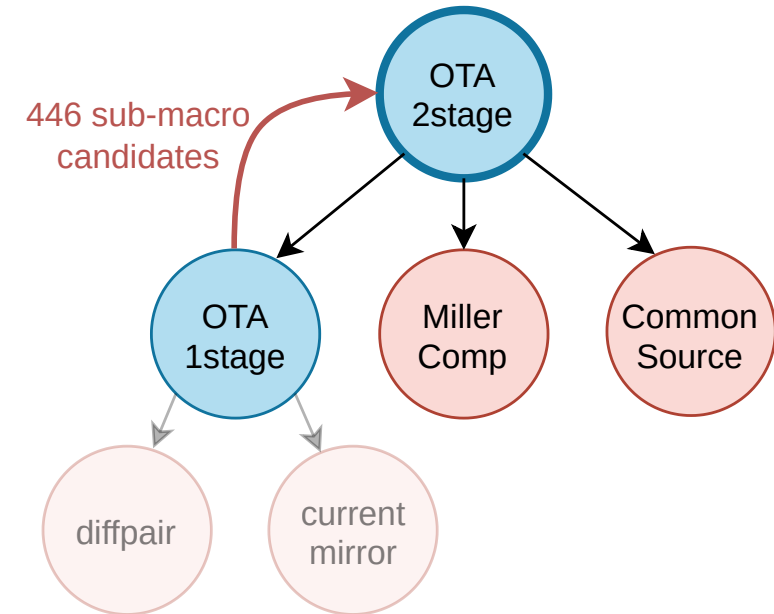
- ▶ TOP exploration: 500k → 2021
 - ▶ Derived constraint: $Gain \geq 28.89$ [dB]
- ▶ Sub-macro exploration: 1050 → 446
 - ▶ Return to parent 446 candidates



ShapelC in Action: 2 Stage Amplifier (OTA)

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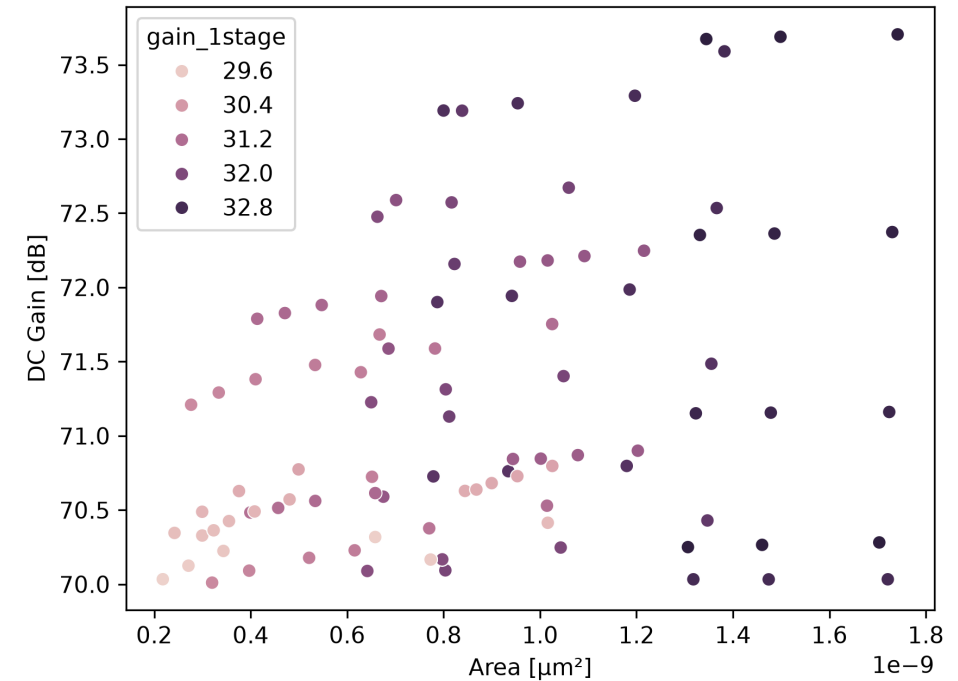
- ▶ TOP exploration: 500k → 2021
 - ▶ Derived constraint: $Gain \geq 28.89$ [dB]
- ▶ Sub-macro exploration: 1050 → 446
 - ▶ Return to parent 446 candidates
- ▶ TOP refresh: 223k → 224



ShapelC in Action: 2 Stage Amplifier (OTA)

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 - ▶ Derived constraint: $Gain \geq 28.89$ [dB]
- ▶ Sub-macro exploration: 1050 → 446
 - ▶ Return to parent 446 candidates
- ▶ TOP refresh: 223k → 224



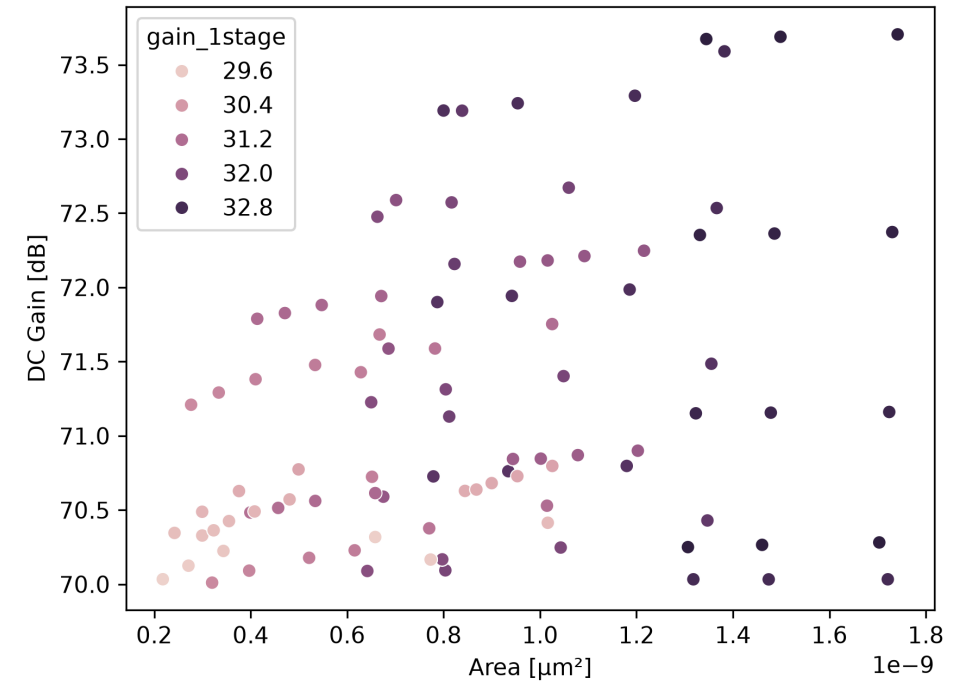
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1. Hierarchical exploration

- ▶ TOP exploration: **500k** → **2021**
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- ▶ Sub-macro exploration: **1050** → **446**
 - ▶ Return to parent 446 candidates
- ▶ TOP refresh: **223k** → **224**

2. Runtime

- ▶ Exploration only: $\approx 2.79\text{s}$
- ▶ Full run incl. LUT load: $\approx 8.38\text{s}$



ShapelC in Action: 2 Stage Amplifier (OTA)

1. Hierarchical exploration

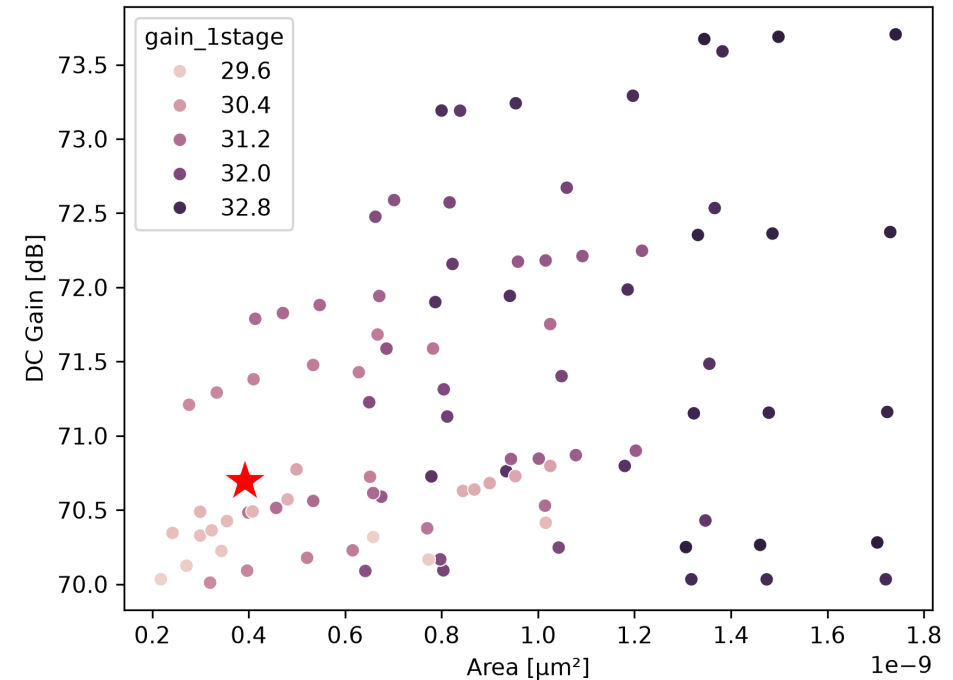
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2. Runtime

- ▶ Exploration only: ≈ 2.79 s
- ▶ Full run incl. LUT load: ≈ 8.38 s

3. Simulation-based optimization

- ▶ Random initialization: no feasible solution found
- ▶ ShapelC-seeded initialization: converged to a feasible solution (★)



Current Status & What is Next ?

► Current status

- ▶ AC analysis supported
- ▶ Hierarchical LUT-based exploration
- ▶ Layout-aware exploration prototype
- ▶ ShapelC-Cellkit integration

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